Smart Control Tower · Architecture & Workflow Walkthrough

Amber Press Shop · Evidence-Linked Operational Review & Deterministic AI Intelligence

APP CALCULATES • AI EXPLAINS

CASE STUDY WALKTHROUGH

Operational Challenge

Plant floor monitors display live DFOS screens, but daily reviews required manual Excel joins across 3 separate reports. Result: unverified OEE metrics (>100%), missing loss reasons, and lack of follow-up accountability.

Engineering Solution

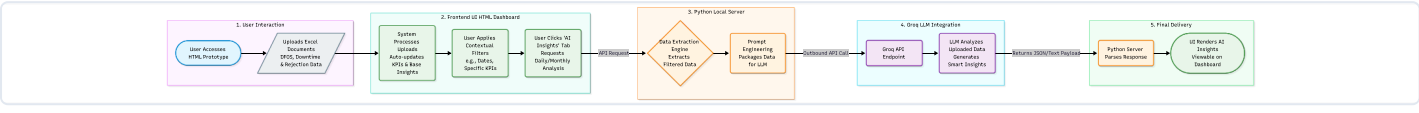
A decoupled Smart Control Tower that unifies multi-report schemas client-side, computes deterministic KPIs & Top 10 Paretos in JS, and routes bounded statistical summaries through a tokenized Python bridge to an LLM chain.

Business & Quality Impact

Review preparation reduced from 90 mins to under 10 seconds. Eliminates AI arithmetic hallucination entirely, protects factory privacy, and validates mathematical reconciliation before management sign-off.

LOGICAL ARCHITECTURE & WORKFLOW PIPELINE

Source: Christmas Shopping Decision-2026-10-12-073041.svg (Mermaid Architecture)



STAGE 1
User Interaction

- **Multi-File Ingestion:** Drag-and-drop Production, Downtime & Rejection reports.
- **Preflight Scoping:** Schema check, date range & machine alias normalization.
- **Context Filters:** Interactive shift, machine cell, model & date slicing.
- **AI Trigger:** On-demand Daily / Monthly briefing dispatch.

STAGE 2
Frontend Dashboard

- **Client Parsing:** In-memory SheetJS parsing (zero server upload).
- **Deterministic KPIs:** Auto-computes Attainment, Availability, Scrap %.
- **Top 10 Paretos:** Ranks loss minutes & reject pieces deterministically.
- **Prompt Packaging:** Minimizes context; excludes sensitive employee remarks.

STAGE 3
Python Local Bridge

- **Local Endpoint:** Pure Python stdlib server binding to `127.0.0.1:4173`.
- **CSRF Token Gate:** 32-byte cryptographically random session bearer token.
- **Payload Throttling:** Enforces <120KB payload limits for cost/latency.
- **Response Parsing:** Normalizes vendor JSON streams into Markdown.

STAGE 4
LLM Resilience Layer

- **Fallback Routing:** Primary OpenAI → Mistral → Groq Edge (`gpt-oss-120b`).
- **Custom User-Agent:** App-specific signature bypassing edge rate limits.
- **Strict Guardrails:** Explains facts; forbidden from recalculating numbers.
- **Zero Retention:** Disables server training with `store: false`.

STAGE 5
Final Delivery

- **DOM Sanitizer:** Renders responsive cards, clean tables & badges.
- **Side-by-Side Proof:** Displays verified Top 10 tables alongside AI narrative.
- **Math Guardrail:** Strips downtime % if Total ≠ Planned + Unplanned.
- **Action Tracking:** RCA case creation with owners & effectiveness checks.

Engineering Safeguards ("Under the Hood")

- **Deterministic Arithmetic:** All sums, rates, and Paretos are computed in JavaScript. The LLM never touches raw arithmetic, guaranteeing 100% mathematical accuracy.
- **Air-Gapped Privacy:** Full spreadsheets and employee remarks remain strictly in the browser. Only bounded statistical summaries are sent to the LLM.
- **Multi-Provider Fallback:** Automatic circuit-breaker failover: OpenAI → Mistral → Groq Edge (handles quota limits or HTTP 429).

The Three Ingested Data Feeds

- **Production Dashboard:** Daily machine output, plan vs actual attainment, baseline operating uptime, and shift scrap count.
- **Detailed Downtime Report:** Event-grain loss intervals, machine ID, planned vs unplanned minutes, and recorded breakdown reasons.
- **Rejection Report:** Defect classifications (Damage, Dent, Hamming NG), reject piece counts, and model scrap ratios.

Production Evolution Roadmap

PHASE 1 • ACTIVE PILOT
Local-First MVP
Client-side XLSX parsing, stdlib server bridge, session storage, and manual Excel drop.

PHASE 2 • NEAR TERM
Hosted Team App
FastAPI + PostgreSQL, Entra ID SSO, immutable batch lineage & durable action register.

PHASE 3 • ENTERPRISE
Automated DFOS Sync
Direct read-only DB view/API, 3-minute IoT micro-sync, and automated anomaly alerts.